

A BOUNDARY-SINGULARITY OBSTRUCTION FOR ROTUND BALL TILINGS

AGENT_LANE_19

SOURCE PROBLEM

The source paper is Carlo Alberto De Bernardi, Tommaso Russo, and Jacopo Somaglia, *Lattice tilings of Hilbert spaces*, arXiv:2505.04267. In Problem 7.5 the authors ask:

Does there exist a rotund and/or smooth normed space X that admits a tiling by balls?

The source paper recalls that separable spaces do not admit tilings by rotund or smooth bounded convex bodies, while in the nonseparable case the known negative results cover LUR or Frechet smooth spaces. The result below is a partial obstruction for arbitrary rotund ball tilings. The final corollary records a diametral strengthening obtained in a later push on the same problem.

DEFINITIONS

A family $\mathcal{T}$ of bodies tiles a normed space X if it covers X and the members have pairwise disjoint interiors. A point $x \in X$ is regular for $\mathcal{T}$ if some neighbourhood of x meets only finitely many members of $\mathcal{T}$; otherwise it is singular.

A boundary point x of a convex body B is a quasi-polyhedral point (QP-point) if there is a neighbourhood U of x such that $[x, y] \subset \partial B$ whenever $y \in U \cap \partial B$. A norm is rotund if its unit sphere contains no non-trivial segment.

A point $x \in S_X$ is locally non-D2 (LND2) if there is $\delta > 0$ such that

$$\text{diam}\{y \in S_X : \|(x + y)/2\| \geq 1 - \delta\} < 2.$$

This condition is weaker than local uniform rotundity at x , but it still rules out diameter-two behaviour in arbitrarily thin midpoint neighbourhoods of x .

IDEA OF THE PROOF

Regularity of a boundary point is a local finiteness condition. Near such a point only finitely many convex tiles are visible, so the tile containing the point is locally cut out by finitely many supporting half-spaces. This is exactly the quasi-polyhedral mechanism used in the source paper. But a quasi-polyhedral point on the boundary of a body in dimension at least two forces a small boundary segment. Therefore a rotund unit ball can never have a regular boundary point in a ball tiling.

The strengthened obstruction uses the opposite case. If a boundary point is singular, then arbitrarily close to it one sees two further unit balls. A standard three-balls lemma converts these three nearby pairwise non-overlapping balls into pairs of unit vectors whose midpoints stay arbitrarily close to the sphere while their mutual distance is arbitrarily close to 2. Thus singularity forces failure of LND2.

Lemma 1 (regular boundary points are QP). *Let $\mathcal{T}$ be a convex tiling of a normed space X . Let $B \in \mathcal{T}$, and let $x \in \partial B$ be regular for $\mathcal{T}$. Then x is a QP-point of B .*

Proof. Since x is regular, there is a neighbourhood U_0 of x that meets only finitely many tiles. Let these tiles be

$$B, B_1, \dots, B_n.$$

Shrinking U_0 , we may assume that

$$U_0 \subset \text{int}(B \cup B_1 \cup \dots \cup B_n).$$

For each k , the convex bodies B and B_k have disjoint interiors and meet at x , so the Hahn–Banach separation theorem gives a functional $f_k \in X^*$ such that

$$\sup f_k(B) = f_k(x) = \inf f_k(B_k).$$

After possibly shrinking to a convex neighbourhood $U \subset U_0$, the local description

$$B \cap U = U \cap \bigcap_{k=1}^n \{z : f_k(z) \leq f_k(x)\}$$

holds. Indeed, if $z \in U \setminus B$, then z lies in one of the adjacent tiles B_k , and the corresponding separating functional excludes z from the right-hand side.

Thus x is locally a finite-half-space boundary point, in particular a QP-point. This is the same local observation recorded as Fact 3.8 in the source paper. $\square$

Lemma 2 (QP points obstruct rotundity). *Let B be the closed unit ball of a normed space X with $\dim X \geq 2$. If ∂B has a QP-point, then B is not rotund.*

Proof. Let $x \in \partial B$ be a QP-point and choose the neighbourhood U from the definition. Since $\dim X \geq 2$ and B is a convex body, ∂B has no isolated

points. Hence there is $y \in U \cap \partial B$ with $y \neq x$. By the QP property, the non-trivial segment $[x, y]$ is contained in ∂B . Therefore the unit sphere contains a non-trivial segment, so B is not rotund. $\square$

Theorem 3 (boundary-singularity obstruction). *Let X be a normed space with $\dim X \geq 2$. Suppose that*

$$\mathcal{T} = \{a_i + B_X : i \in I\}$$

is a tiling of X by translates of the closed unit ball. If the norm of X is rotund, then every boundary point of every tile is singular for $\mathcal{T}$.

Proof. It suffices to consider the tile B_X , since translations preserve both regularity and rotundity. Suppose that $x \in \partial B_X$ is regular for $\mathcal{T}$. By Lemma 1, x is a QP-point of B_X . By Lemma 2, B_X is not rotund, contradicting the assumption. Thus no boundary point can be regular. $\square$

Corollary 4. *No normed space of dimension at least two with a rotund norm admits a locally finite tiling by translates of its closed unit ball.*

Proof. In a locally finite tiling every point is regular. This contradicts Theorem 3, because the unit ball has nonempty boundary. $\square$

Lemma 5 (three nearby balls force local diameter two). *Let X be a normed space, let $B_0 = B_X$, and let B_1, B_2 be closed balls of radius one whose interiors are pairwise disjoint from each other and from $\text{int } B_0$. Suppose that $y_i \in \partial B_i$ for $i = 0, 1, 2$, and that $\text{diam}\{y_0, y_1, y_2\} \leq \eta$. Then*

$$\text{diam}\{y \in S_X : \|y_0 + y\| \geq 2 - \eta\} \geq 2 - 2\eta.$$

Proof. This is the standard three-balls estimate of De Bernardi–Somaglia–Vesely [2, Lemma 3.7] and De Bernardi–Vesely [3, Lemma 4.5]. It is also quoted in the packing-constant literature in exactly this form. We use it only as a local geometric lemma. $\square$

Lemma 6 (singular boundary points fail LND2). *Let $\mathcal{T} = \{a_i + B_X : i \in I\}$ be a tiling by translates of the closed unit ball. If $x \in S_X$ is singular for $\mathcal{T}$, then for every $\delta > 0$,*

$$\text{diam}\{y \in S_X : \|(x + y)/2\| \geq 1 - \delta\} = 2.$$

In particular, x is not an LND2 point.

Proof. Fix $\eta > 0$. Since x is singular, every neighbourhood of x meets infinitely many tiles. Choose two distinct non-central tiles T_1, T_2 that meet $B(x, \eta)$. For $i = 1, 2$, choose $z_i \in T_i \cap B(x, \eta)$.

The point x is not an interior point of T_i : otherwise a small neighbourhood of x inside T_i would meet $\text{int } B_X$, contradicting the disjointness of tile interiors. Hence the segment from z_i to x contains a point $y_i \in \partial T_i$ with $\|y_i - x\| \leq \eta$. Taking $y_0 = x \in \partial B_X$, the three unit balls B_X, T_1, T_2 satisfy the hypotheses of Lemma 5 with $\text{diam}\{y_0, y_1, y_2\} \leq 2\eta$. Therefore

$$\text{diam}\{y \in S_X : \|x + y\| \geq 2 - 2\eta\} \geq 2 - 4\eta.$$

Equivalently,

$$\text{diam}\{y \in S_X : \|(x + y)/2\| \geq 1 - \eta\} \geq 2 - 4\eta.$$

For a fixed $\delta > 0$, let $\eta < \delta$ tend to 0. The displayed sets increase when the threshold is weakened, so the set corresponding to δ has diameter 2. Thus x is not LND2. $\square$

Corollary 7 (diametral strengthening). *Let X be a normed space with $\dim X \geq 2$. If translates of B_X tile X and the norm of X is rotund, then no point of S_X is LND2.*

Proof. By Theorem 3, every point of $S_X = \partial B_X$ is singular for the tiling. Lemma 6 then applies at every point of the unit sphere. $\square$

Corollary 8 (smooth no-flat variant). *Let X admit a tiling by translates of its closed unit ball. If a boundary point x of a tile is regular and the norm is Gateaux smooth at x , then x is a flat point of the unit ball. Consequently, a Gateaux smooth ball tiling with no flat boundary points must have every boundary point singular.*

Proof. The regular point is QP by Lemma 1. The standard local convexity observation recalled in the source paper says that every Gateaux smooth QP-point is flat: the unique supporting hyperplane at a smooth quasi-polyhedral point must contain a relative neighbourhood of the boundary. The final statement is immediate. $\square$

LIMITATIONS AND NOVELTY CHECK

This is a partial obstruction, not a solution to Problem 7.5. It leaves open whether a rotund or smooth normed space with a necessarily everywhere-singular ball tiling exists. The strengthened conclusion shows that a positive rotund example would have to be extreme in a second way: every unit vector must fail LND2, so the unit sphere must be rotund but locally diameter-two at every point.

I checked the run indexes for arXiv:2505.04267 and the phrases *lattice tilings*, *rotund*, *smooth*, and *ball tiling*; no duplicate packet appeared. A bounded web search found the source arXiv page and no later exact resolution of Problem 7.5. The argument intentionally uses the source paper’s local QP facts, so human review should compare the statement against the older Klee–Maluta–Zanco and Klee–Tricot literature.

REFERENCES

- [1] C. A. De Bernardi, T. Russo, and J. Somaglia, *Lattice tilings of Hilbert spaces*, arXiv:2505.04267.
- [2] C. A. De Bernardi, J. Somaglia, and L. Vesely, *Star-finite coverings of Banach spaces*, J. Math. Anal. Appl. 491 (2020), 124384.
- [3] C. A. De Bernardi and L. Vesely, *Tilings of normed spaces*, Canad. J. Math. 69 (2017), 321–337.

- [4] V. Klee, E. Maluta, and C. Zanco, *Tiling with smooth and rotund tiles*, Fund. Math. 126 (1986), 269–290.
- [5] V. Klee and C. Tricot, *Locally countable plump tilings are flat*, Math. Ann. 277 (1987), 315–325.